Xiang Chen

Academic Building 3664, Lifts 31/32, Clear Water Bay, Kowloon, Hong Kong

xchenht@cse.ust.hk · xchenht.student.ust.hk · x14ngch3n · 0009-0007-0626-6888 · Google Scholar

Education

The Hong Kong University of Science and Technology 2024/08 - now

Ph.D. student in Prism Lab, CSE, supervised by Charles Zhang.

Shanghai Jiao Tong University 2021/09 - 2024/03

Master degree in Cyber Security, supervised by Yue Wu and Jiaping Gui.
Thesis: C/C++ system software Static analysis techniques through the lens of Integer Overflow Detection

Shanghai Jiao Tong University 2017/09 - 2021/06

Bachelor degree in Information Security, selected to the Zhiyuan Honor Program.
Thesis: Vulnerability Detection and Analysis for Massive Large-scale IoT Devices

Peking University Summer School 2019/07 - 2019/08

Coordinator of five selected granted students from Shanghai Jiao Tong University.

Publications

- (In submission) **Xiang Chen**, Anshunkang Zhou, Charles Zhang. ShadowTV: Scalable Translation Validation of Binary Lifters Anchored in Compiler Debug Information.
- (In submission) **Xiang Chen**, Chengfeng Ye, Anshunkang Zhou, Charles Zhang. IntEcho: Detecting Recurring Integer Overflow Vulnerabilities in COTS Binaries.
- (In submission) **Xiang Chen**, Anshunkang Zhou, Chengfeng Ye, Charles Zhang. ORCA: Demand-Driven Program Analysis for Grounded Agentic Binary Reasoning.
- **Xiang Chen**, Anshunkang Zhou, Chengfeng Ye, Charles Zhang. ClearAgent: Agentic Binary Analysis for Effective Vulnerability Detection. The 1st International Workshop on Language Models and Programming Languages (**LMPL '25**)
- **Xiang Chen**. IntTracer: Sanitization-aware IO2BO Vulnerability Detection across Codebases. In 2024 IEEE/ACM 46th International Conference on Software Engineering, Student Research Competition (**ICSE SRC '24**)
- Tianming Zheng, Haojun Liu, Hang Xu, **Xiang Chen**, Ping Yi, Yue Wu, Few-VulD: A Few-shot learning framework for software vulnerability detection, Computers & Security, 2024

Services

Artifact Evaluation: USENIX Sec / PLDI / OSDI / CCS '25, NDSS / IEEE S&P / PLDI '26 2025/02 - now

Sub-Reviewer: TSE '25, LLM4Code / OOPSLA / CCS '26, ICSE '27 2025/04 - now

China Computer Federation (CCF) Student Chapter in SJTU, Executive Committee 2022/11 - 2023/12

GeekPwn 2020 volunteer 2020/10

Awards

HKUST Research Travel Grant (RTG) 2025/10

HKUST Postgraduate Scholarship (PGS) 2024/09 - now

Shanghai Jiao Tong University Outstanding Graduate **(<10%)** 2024/03

Rong Chang Leadership Scholarship (<1%) 2021/11 - 2023/11

DEFCON CTF 30 **2nd** place (played with Katzebin) 2022/08

Zhiyuan Honor Bachelor Degree **(Cum Laude, <1%)** 2021/06

Shanghai Outstanding Graduate **(<5%)** 2021/06

Teaching Experience

HKUST Firebird CTF Team Coach 2025/08 - now

Provide training on reverse engineering and binary exploitation. Organizing firebird CTF contests.

COMP 3021: Java Programming 2025/01 - 2025/06

Design course project: an LLM chat client that supports configurable persistence, large-scale functional, and parallel querying.

IS308: Computer System Security (The first "John Hopcroft" Class) 2023/02 - 2023/06

Mentoring five labs in binary/web security and cryptography. Hosting a CTF-style final exam.

NIS7021: Software and System Security 2022/10 - 2023/01

Design two labs in reverse engineering and binary hardening using LLVM.

Industry Experience

NIO Inc. 2022/10 - 2023/10

Funding project "decreasing FP and FN rates in static C/C++ program analysis" from Cyber Security Academy Student Innovation Grant Program. The project focuses on using Facebook Infer's Abstract Interpretation framework and taint analysis technique in detecting Uninitialized Value bugs in Linux Kernel.

Shanghai Qizhi Institute 2022/07 - 2022/11

G.O.S.S.I.P Research Group Internship, weekly paper reading and research on (1) automatic program repair using LLVM Instrumentation and the Daikon invariant detector and (2) automatic bug fix for use-after-move issues in C++ 11 using Clang-Tidy.

Shanghai Feysh Technology Co.,Ltd 2021/07 - 2021/09

Manually review more than 4000 analysis results of ClangStaticAnalyzer on Juliet C/C++ Test Suite. Implement four taint-style bug checkers for SEI CERT C Coding Standard.

Open-Source Contributions

Open Source Promotion Plan (openEuler) 2023/07 - 2023/09

Enhance LLVM InstCombine pass with a peephole optimization, which can eliminate `abs()` in ternary expressions like: `x>y? abs(x-y+1):0` and combine the original if-else-branch to linear CFG using the AArch64 `csinc` instruction.

SJTUBeamer 2021/04 - 2021/11

The Shanghai Jiao Tong University official LaTeX beamer template with 800+ stars, made during Bachelor's thesis.

Bug Hunting

- see https://x14ngch3n.github.io/bugs/ for details.

Talks

- **Xiang Chen**, Siqi Ma. 2023. Custom Memory Functions Demystified: A tutorial of memory corruption detection using Goshawk. In ACM ASIA Conference on Computer and Communications Security (**ASIA CCS '23**)
- **Xiang Chen**. 2023. C/C++ static analysis with LLVM compiler infrastructure. Voice of Information Security-Young

Skills

- Programming Languages: C/C++, Python > Java, Rust, OCaml
- Development Toolchains: Claude Code/Codex/Gemini CLI, VSCode, CMake, LLVM/Clang, GDB, Docker, Git
- Capture-The-Flag: IDA/Binja/Angr, Pwntools, Wireshark, Sage